



QP CODE: 23709211



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Reg No :

Name :

M.C.A. DEGREE EXAMINATION, MAY 2023

First Semester

Master of Computer Application

**Core - MCACT101 - MATHEMATICAL & STATISTICAL FOUNDATION FOR
COMPUTER APPLICATIONS**

2020 Admission Onwards

98B2973A

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. If A and B are any two sets, prove that $A \cup (B - A) = A \cup B$
2. Define equivalence class
3. Discuss one -one and onto functions with examples
4. What is Conjunction? Construct the truth table for $\sim p \wedge (p \rightarrow q)$
5. Explain the rules of inference in propositional calculus
6. Express the following sentence in symbolic form using quantifiers – “ All the world loves a lover”.
7. Distinguish between correlation and regression.
8. The probability that a student passes mathematics is $\frac{2}{3}$, the probability that he passes statistics is $\frac{4}{9}$. If the probability of passing at least one subject is $\frac{4}{5}$, what is the probability that he will pass both the subjects?
9. Obtain the binomial distribution for which mean is 10 and the variance is 5.
10. What is statistical Inference? What are the branches of statistical Inference?
11. What is one tailed and two tailed test
12. Define test statistic. Write the test statistic in large sample test for single mean

(10×3=30 marks)





Part B

Answer **all** questions

Each question carries **9** marks

13. a) Define equivalence relation. Find the equivalence relation that generates the partition $\{0,1,3\}$, $\{2\}$ and $\{4\}$ of the set $A=\{0,1,2,3,4\}$. Also write relation matrix and draw the graph of the relation

OR

- b) Let f and g are function from R to R defined by $f(x)=2x+1$ and $g(x)=x^2-2$; $\forall x$ respectively. Then find f^{-1} , $f \circ g$, $g \circ f$, and $f \circ f$.

14. a) Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $(P \vee Q)$, $Q \rightarrow R$, $P \rightarrow M$ and $\sim M$

OR

- b) Prove the implication $\forall x (P(x) \rightarrow Q(x))$, $\forall x (R(x) \rightarrow \sim Q(x)) \Rightarrow \forall x (R(x) \rightarrow \sim P(x))$

15. a) Obtain the Rank Correlation Coefficient for the following data.

X	17	13	15	16	6	11	14	9	7	12
Y	36	46	35	24	12	18	27	22	2	8

OR

- b) Three identical boxes contain two balls each. One has both red, one has one red and one black and the third has two black balls. A person chooses a box at random and takes out a ball. i) Find the probability that the selected ball is red ii) If the ball is red find the probability that the other ball in the box is also red

16. a) The number of accidents in an year attributed to taxi drivers in a city follows Poisson distribution with mean 3. Out of 1000 taxi drivers find approximately the number of drivers with (i) No accident in a year (ii) more than 3 accidents in a year

OR

- b) In a Binomial distribution consisting of 5 independent trials, the first and second terms are 0.4096 and 0.2048 respectively. Find the parameter 'p'.

17. a) A stenographer claims that she can take dictation at the rate of 120 words per minute. Can we reject her claim on the basis of 100 trials in which she demonstrates a mean of 116 words with standard deviation of 15 words? Use 5% level of significance

OR





- b) The theory predicts the proportion of beans , in the four groups A, B,C, and D should be 9:3:3:1. In an experiment among 1,600 beans the numbers in the four groups were 882, 313, 287 and 118. Does the experimental result support the theory? (The table value of for 3 d.f. at 5% level of significance is 7.81).

(5×9=45 marks)

